MIL-R-23417/33A
28 APRIL 1970
SUPERSEDING
MIL-R-23417/33
16 October 1967

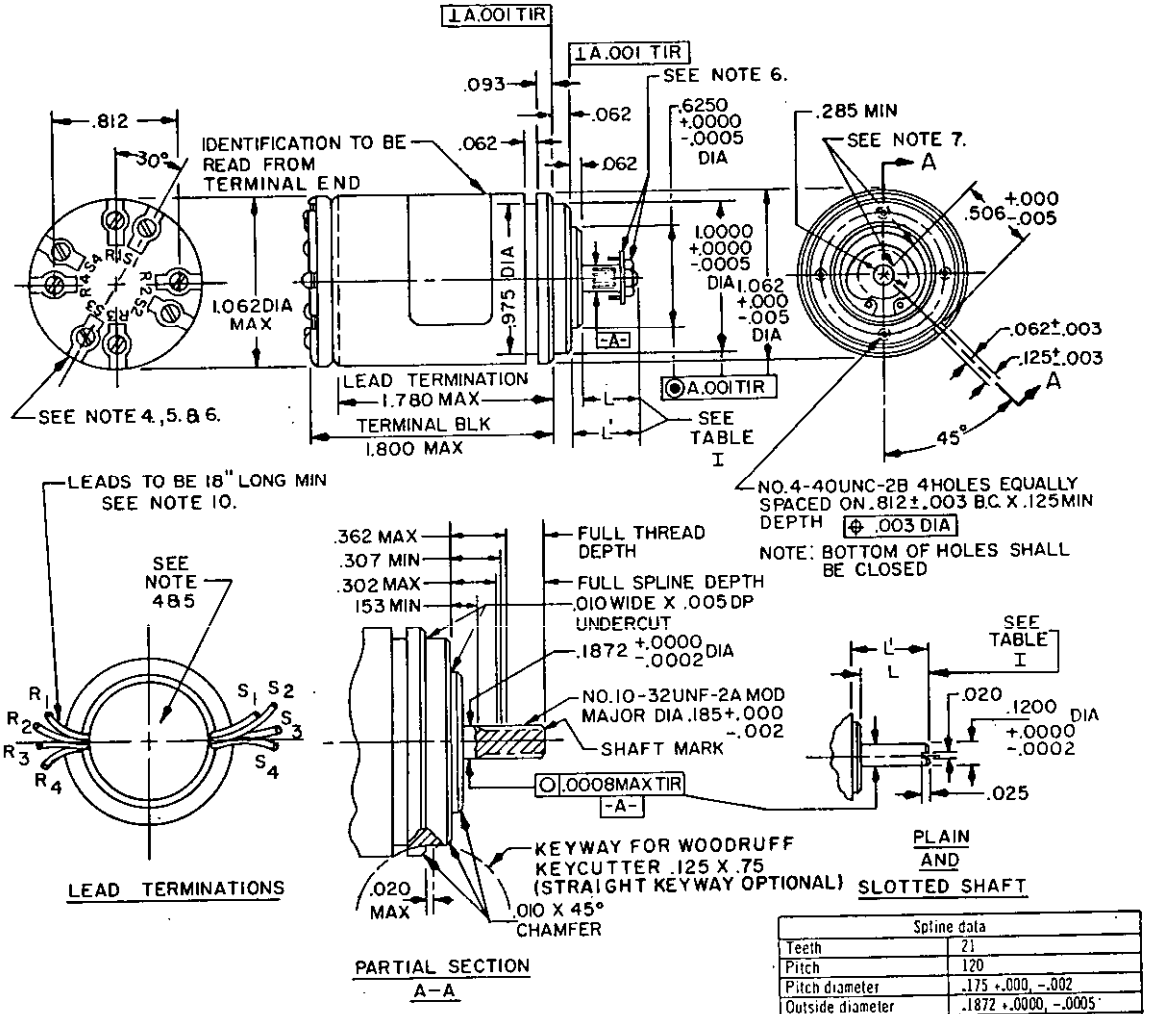
MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL

TYPE 11R21N4B

The complete requirements for procuring the resolver described herein shall consist of this document and the latest issue of MIL-R-23417.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Spline data	
Teeth	21
Pitch	120
Pitch diameter	.175 +.000, -.002
Outside diameter	.1872 +.0000, -.0005
Pressure angle	20°
Root diameter maximum	.155
Tooth form	Full depth involute

Note: The end function of this spline is for positive rotational fastenings. It is not to be used as a gear. This spline shall be inspected by (spline) go composite ring gage MS 90406

FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are .XXX ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Terminal block designed for MS17182-2 terminal lug (an approved 12-terminal block may be used provided all unused terminal positions are closed). Terminal lead wires shall depart through the back surface of the terminal end of the resolver in any acceptable manner.
5. Minor variations of unit configuration are permitted with undimensioned design detail.
6. The following list of hardware shall be packaged in a ventilated envelope made of material having a pH value of 6-8 and compatible with the weight of the contents. The packaged hardware shall be placed with the unit resolver as follows:

Machine screw	MS 35276-203	8 Required (*) (**)
Lockwasher	MS 35338-134	8 Required (*) (**)
Drivewasher	MS 17186-6	1 Required (*)
Nut	MS 17187-2	1 Required (*)

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal board termination.

7. Mark on housing is to match shaft mark within 10° when rotor is set at approximately resolver zero, and will bear no relation to the center lines shown on the drawing.
8. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-R-23417.
9. For information only see synchro mounting clamp MS 17183 or equal.
10. Leads shall be type E-28.
11. A voltage value of .5 to 26 volts shall be entered in the Volts designation space on the identification plate.
12. The weight shall be attached to the end of each wire, pulling straight away from the resolver. The lead shall then be bent 90° at the point of exit from the housing and while in this attitude the resolver rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or permanent deformation of the insulation shall be cause for rejection under qualification tests.

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RESOLVER, ELECTRICAL: TYPE 11R21N4B

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

Requirements	Value	Unit	Tolerance
1. Excitation winding	Stator	---	---
2. Rated frequency	400	Hz	±1%
3. Input voltage:	11.8	Volts rms	±1%
Test	11.8	Volts rms	±1%
4. Primary current	6.0	Milliamps	Nominal
5. Primary power	0.011	Watts	Nominal
6. DC resistance:			
Rotor	1200	Ohms	Reference
Stator	170	Ohms	Reference
7. Impedance:			
Zro	1760 + j7530	Ohms	R±15%, +jx±10%
Zso	330 + j1980	Ohms	R±15%, +jx±10%
Zrs	1790 + j670	Ohms	Reference
8. Transformation ratio Output/input	1.902	---	±2%
9. Equality of transformation ratio Output/input	±0.1	Percent	---
10. Phase shift Output/input	5.2 (Lead)	Degrees	±1.0
11. Accuracy Electrical error	±3.0	Minutes	Maximum
12. Null voltage:			
Fundamental	15	Mv rms	Maximum
Total	20	Mv rms	Maximum
13. Shift of resolver zero With frequency	1	Minutes	Maximum
14. Temperature rise	5	°C	Maximum
15. Friction torque:			
@ 23°C	0.07	Oz-In	Maximum
@ -55°C and +125°C	0.15	Oz-In	Maximum
16. Radial play	0.0006 (For reversal of 1/2 lb load)	Inches	Maximum
17. End play	0.0001 - 0.0015 (For reversal of 1 lb load)	Inches	Min-Max
18. Wire lead stress (see note 12)	2	Pound	Minimum

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

M23417/ 33 -01

Basic document number	Spec Sheet number	Dash number from table I

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TABLE I. Dash numbers and characteristics.

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				L (ref)	L' - dim to first mounting surface (±.010)
-01	.187	Spline	Terminal board	.500	.562
-07	.120	Plain	Wire leads	.312	.374
-08	.120	Plain	Wire leads	.375	.437
-09	.120	Plain	Wire leads	.437	.499
-10	.120	Plain	Wire leads	.500	.562
-11	.120	Plain	Wire leads	.562	.624

Custodians:
Army - MU
Navy - AS
Air Force - 80

Review activities:
Army - MU, EL
Navy - AS, SH, EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MI, ME, AV
Navy - CG, SH
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5990-0249)

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